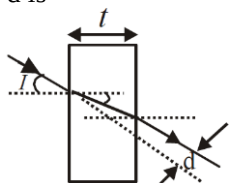


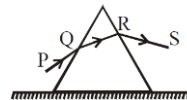
PHYSICS

- A man runs toward a mirror at a speed 15 m/s. The speed of the image relative to the man is
 (1) 15 ms^{-1} (2) 30 ms^{-1}
 (3) 35 ms^{-1} (4) 20 ms^{-1}
- A man of length h requires a mirror, to see his own complete image of length at least equal to
 (1) $h/4$ (2) $h/3$ (3) $h/2$ (4) h
- A ray of light is incident on a thick slab of glass of thickness t as shown in the figure. The emergent ray is parallel to the incident ray but distance d . If the angles are small then d is



- (1) $t \left(1 - \frac{i}{r}\right)$ (2) $rt \left(1 - \frac{i}{r}\right)$
 (3) $it \left(1 - \frac{r}{i}\right)$ (4) $t \left(1 - \frac{r}{i}\right)$
- The graph between angle of deviation (δ) and angle of incidence (i) for a triangular prism is represented by
 (1) (2)
 (3) (4)
- Two slits, 4 mm apart, are illuminated with light of wavelength $6000 \text{ \AA}$. What will be the fringe width on a screen placed 2 m far from the slits?
 (1) 0.12 mm (2) 0.3 mm
 (3) 3.0 mm (4) 4.0 mm
- A concave mirror is used to focus the image of a flower on a nearby wall 120 cm from the flower. If a lateral magnification of 16 is desired, the distance of the flower from the mirror should be
 (1) 8 cm (2) 12 cm (3) 80 cm (4) 120 cm
- An object is placed 40 cm from a concave mirror of focal length 20 cm. The image formed is
 (1) Real, inverted and small in size
 (2) Real, inverted and smaller
 (3) Virtual, erect and larger
 (4) Virtual, erect and smaller

- A beam of monochromatic blue light of wavelength $4200 \text{ \AA}$ in air travels in water ($\mu = 4/3$). Its wavelength in water will be
 (1) $2800 \text{ \AA}$ (2) $5600 \text{ \AA}$
 (3) $3150 \text{ \AA}$ (4) $4000 \text{ \AA}$
- An equilateral prism is placed on a horizontal surface. A ray PQ is incident onto it. For minimum deviation



- (1) PQ is horizontal
 (2) QR is horizontal
 (3) RS is horizontal
 (4) Any one will be horizontal
- A convex mirror has a focal length f . A real object is placed at a distance f in front of it from the pole produces an image at
 (1) Infinity (2) f
 (3) $f/2$ (4) $2f$
- Under which of the following conditions will a convex mirror of focal length f produce an image that is erect, diminished and virtual
 (1) Only when $2f > u > f$
 (2) Only when $u = f$
 (3) Only when $u < f$
 (4) Always
- For the same angle of incidence, the angles of refraction in media 'P', 'Q', 'R' and 'S' are 50° , 40° , 30° , 20° respectively. The speed of light is minimum in medium
 (1) P (2) Q (3) R (4) S
- In Young's double slit interference pattern the fringe width
 (1) Can be changed only by changing the wavelength of incident light
 (2) Can be changed only by changing the separation between the two slits
 (3) Can be changed either by changing the wavelength or by changing the separation between the two slits
 (4) Is a universal, hence cannot be changed
- A convex lens of focal length 40 cm is in contact with a concave lens of focal length 25 cm. The power of combination is
 (1) -1.5 D (2) -6.5 D
 (3) $+6.5 \text{ D}$ (4) $+6.67 \text{ D}$
- Monochromatic light is refracted from air into the glass of refractive index μ . The ratio of the wavelength of incident and refracted waves is
 (1) $1 : \mu$ (2) $1 : \mu^2$ (3) $\mu : 1$ (4) $1 : 1$

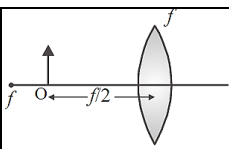
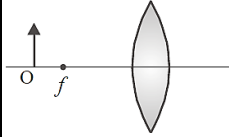
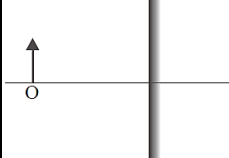
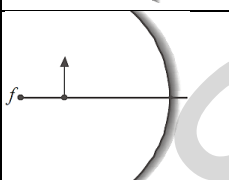
16. A light is traveling from air into a medium. Velocity of light in a medium is reduced to 0.75 times the velocity in air. Assume that angle of incidence 'i' is very small, the deviation of the ray is

(1) i (2) $\frac{i}{3}$ (3) $\frac{i}{4}$ (4) $\frac{3i}{4}$

17. In young's double slit experiment, the wavelength of red lights is 7.8×10^{-5} cm and that of blue light 5.2×10^{-5} cm. The value of n for which $(n + 1)^{\text{th}}$ blue bright band coincides with n^{th} red band is

(1) 4 (2) 3 (3) 2 (4) 1

18. Match the following Column II gives nature of image formed in various cases given in Column I

Column I		Column II	
(A)		(1)	Real
(B)		(2)	Inverted
(C)		(3)	Virtual
(D)		(4)	Upright
		(5)	Magnified

(1) (A)→(1,2,3); (B)→(4,5); (C)→(2); (D)→(1, 5)
 (2) (A)→(3,4,5); (B)→(1,2,5); (C)→(3,4); (D)→(3,4,5)
 (3) (A)→(1); (B)→(2, 3); (C)→(4, 5); (D)→(1)
 (4) (A)→(4); (B)→(3, 4); (C)→(2, 3); (D)→(1)

19. A spherical surface of radius of curvature R separates air (refractive index 1.0) from glass (refractive index 1.5). The centre of curvature is in the glass. A point object P placed in air is found to have a real image Q in the glass. The line PQ cuts the surface at a point O and $PO = OQ$. The distance PO is equal to

(1) 5 R (2) 3 R (3) 2 R (4) 1.5 R

20. In a Young's double slit experiment, the intensity at a point where the path difference is $\frac{\lambda}{6}$ (λ being the wavelength of light used) is I. If I_0 denotes the maximum intensity, $\frac{I}{I_0}$ is equal to

(1) $\frac{3}{4}$ (2) $\frac{1}{\sqrt{2}}$ (3) $\frac{\sqrt{3}}{2}$ (4) $\frac{1}{2}$

21. When light is reflected from air into glass
 (1) Its wavelength and frequency both increase
 (2) Its wavelength increases but frequency remains unchanged
 (3) Its wavelength decreases but frequency remains unchanged
 (4) Its wavelength and frequency both decrease

22. The wavelength of light in air and some other medium are respectively λ_a and λ_m . The refractive index of medium is

(1) λ_a/λ_m (2) λ_m/λ_a
 (3) $\lambda_a \times \lambda_m$ (4) None of these

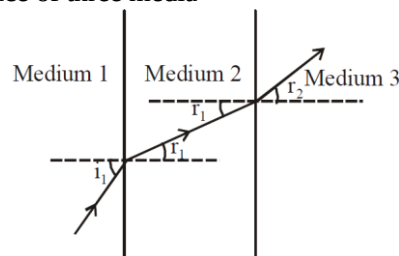
23. Four lenses are made from the same type of glass. The radius of curvature of each is given below. Which will have the greatest positive power

(1) 10 cm convex and 15 cm concave
 (2) 5 cm convex and 10 cm concave
 (3) 15 cm convex and plane
 (4) 20 cm convex and 30 cm concave

24. In a two slit experiment with monochromatic light, fringes are obtained on a screen placed at some distance from the plane of slits. If the screen is moved by 5×10^{-2} m towards the slits, the change in fringe width is 3×10^{-5} m. If the distance between slits is 10^{-3} m, the wavelength of light will be

(1) 3000 Å (2) 4000 Å
 (3) 6000 Å (4) 7000 Å

25. The following figure shows refraction of light at the interface of three media



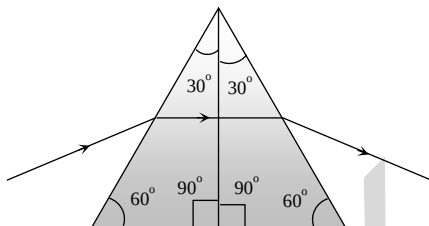
Correct the order of optical density (d) of the media is

(1) $d_1 > d_2 > d_3$ (2) $d_2 > d_1 > d_3$
 (3) $d_1 > d_3 > d_2$ (4) $d_2 > d_3 > d_1$

26. All of the following statements are correct except
- (1) the magnification produced by a convex mirror is always less than one
 - (2) a virtual, erect, same-sized image can be obtained using a plane mirror
 - (3) a virtual, erect, magnified image can be formed using a concave mirror
 - (4) a real, inverted, same-sized image can be formed using a convex mirror.

27. The sun's diameter is $1.4 \times 10^9 m$ and its distance from the earth is $10^{11} m$. The diameter of its image, formed by a convex lens of focal length 2m will be
- (1) 0.7 cm
 - (2) 1.4 cm
 - (3) 2.8 cm
 - (4) Zero (i.e. point image)

28. Two identical prisms 1 and 2, each will angles of 30° , 60° and 90° are placed in contact as shown in figure. A ray of light passed through the combination in the position of minimum deviation and suffers a deviation of 30° . If the prism 2 is removed, then the angle of deviation of the same ray is



- (1) Equal to 15°
 - (2) Smaller than 30°
 - (3) More than 15°
 - (4) Equal to 30°
29. A light has amplitude A and angle between analyser and polarizer is 60° . Light is reflected by analyser has amplitude
- (1) $A\sqrt{2}$
 - (2) $A/\sqrt{2}$
 - (3) $\sqrt{3}A/2$
 - (4) $A/2$
30. When light is incident on a doubly refracting crystal, two refracted rays-ordinary ray (O-ray) and extra ordinary ray (E-ray) are produced. Then
- (1) Both O-ray and E-ray are polarized perpendicular to the plane of incidence
 - (2) Both O-ray and E-ray are polarized in the plane of incidence
 - (3) E-ray is polarized perpendicular to the plane of incidence and O-ray in the plane of incidence
 - (4) E-ray is polarized in the plane of incidence and O-ray perpendicular to the plane of incidence

Direction (Q.31 & Q.32) : The following questions consist of two statements each, printed as **Assertion** and **Reason**.

- (A) If both, Assertion and Reason are true and the Reason is the correct explanation of the Assertion.
- (B) If both, Assertion and Reason are true but Reason is not a correct explanation of the Assertion.
- (C) If Assertion is true but the Reason is false.
- (D) If both, Assertion and Reason are false.

31. **Assertion:** A prism is the source of colours of light.
Reason: A prism has same refractive index for different colors of light.

(1) A (2) B (3) C (4) D

32. **Assertion:** No diffraction is produced in sound waves near a very small opening.

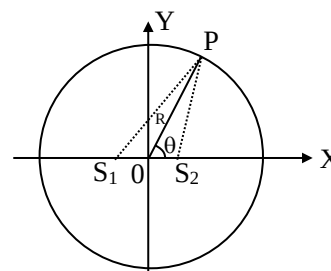
Reason: For diffraction to take place, the aperture of opening should be of same order as wavelength of waves.

(1) A (2) B (3) C (4) D

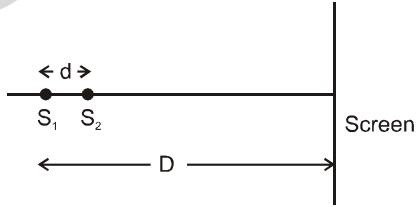
33. A convex mirror of focal length f forms an image which is $\frac{1}{n}$ times the object. The distance of the object from the mirror is

(1) $(n-1)f$ (2) $\left(\frac{n-1}{n}\right)f$
 (3) $\left(\frac{n+1}{n}\right)f$ (4) $(n+1)f$

34. Two coherent sources of light S_1 and S_2 , equidistant from the origin, are separated by a distance 2λ as shown. They emitted light of wavelength λ . Interference is observed on a screen placed along the circle of large radius R . Point is seen to be a point of constructive interference. Then angle θ (other than 0° and 90°) is

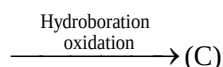
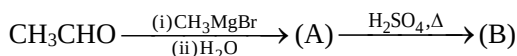


- (1) 45°
- (2) 30°
- (3) 60°
- (4) Not possible in the first quadrant

35. If an interference pattern has maximum and minimum intensity in the ratio of 36 : 1, then what will be the ratio of amplitudes?
(1) 5 : 7 (2) 7 : 4 (3) 4 : 7 (4) 7 : 5
36. In the interference pattern, energy is
(1) Created at the position of maxima
(2) Destroyed at the position of minima
(3) Conserved but is redistributed
(4) None of these
37. When light travels from one medium to the other of which the refractive index is different, then which of the following will change
(1) Frequency, wavelength and velocity
(2) Frequency and wavelength
(3) Frequency and velocity
(4) Wavelength and velocity
38. A slit of width a is illuminated by parallel monochromatic light of wavelength λ . The value of a at which the first minimum of the diffraction pattern will form at $\theta = 30^\circ$ is
(1) $\lambda/2$ (2) λ (3) 2λ (4) 3λ
39. Which of the following is not a correct statement
(1) The wavelength of red light is greater than the wavelength of green light
(2) The wavelength of blue light is smaller than the wavelength of orange light
(3) The frequency of green light is greater than the frequency of blue light
(4) The frequency of violet light is greater than the frequency of blue light
40. A ray of light is incident at an angle i on a glass slab of refractive index μ . The angle between reflected and refracted light is 90° . Then the relationship between i and μ is
(1) $i = \tan^{-1}\left(\frac{1}{\mu}\right)$ (2) $\tan i = \mu$
- (3) $\sin i = \mu$ (4) $\cos i = \mu$
41. A ray of light is incident on the surface of a glass plate at an angle of incidence equal to Brewster's angle ϕ . If μ represents the refractive index of glass with respect to air, then the angle between the reflected and the refracted rays is
(1) $90^\circ + \phi$ (2) $\sin^{-1}(\mu \cos \phi)$
(3) 90° (4) $90^\circ - \sin^{-1}\left(\frac{\sin \phi}{\mu}\right)$
42. An object is placed at 15 cm in front of a concave mirror whose focal length is 10 cm. The image formed will be
(1) Magnified and inverted
(2) Magnified and erect
(3) Reduced in size and erect
(4) Increased in size and erect
43. Two coherent point sources S_1 and S_2 are separated by a small distance 'd' as shown. The fringes obtained on the screen will be :

(1) points (2) straight lines
(3) semi-circles (4) concentric circles
44. Two monochromatic light waves of amplitudes A and $2A$ interfering at a point have a phase difference of 60° . The intensity at that point will be proportional to
(1) $3A^2$ (2) $5A^2$
(3) $7A^2$ (4) $9A^2$
45. In Young's double slit experiment, the fringes are displaced by a distance x when a glass plate of refractive index 1.5 is introduced in the path of one of the beams. When this plate is replaced by another plate of the same thickness, the shift of fringes is $(3/2)x$. The refractive index of the second plate is
(1) 1.75 (2) 1.50 (3) 1.25 (4) 1.00

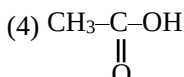
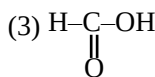
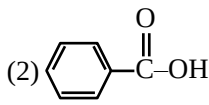
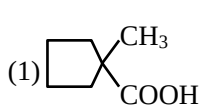
CHEMISTRY

46. Compounds A and C in the following reaction are _____.

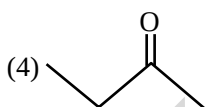
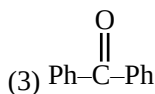
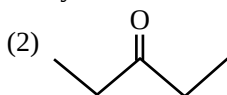
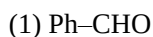


- (1) identical
(2) positional isomers
(3) functional isomers
(4) optical isomer

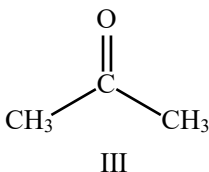
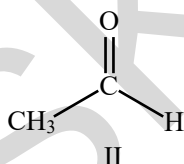
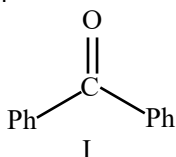
47. Which of the following give HVZ reaction –



48. Compound 'A' give positive test with 2,4-DNP and with I_2/NaOH . Compound 'A' may be:



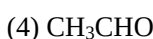
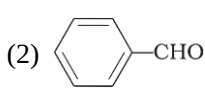
49. The correct order of reactivity of the carbonyl compound



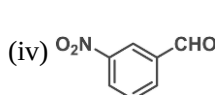
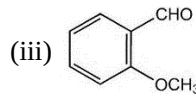
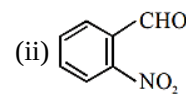
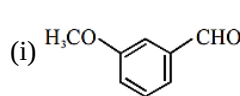
With respect to their addition reactions with PhMgBr is

- (1) $\text{I} > \text{II} > \text{III}$ (2) $\text{III} > \text{II} > \text{I}$
(3) $\text{II} > \text{III} > \text{I}$ (4) $\text{I} > \text{III} > \text{II}$

50. Cannizaro's reaction is not given by _____.

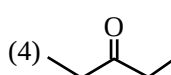
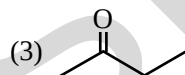
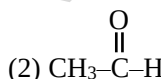


51. The increasing order of the following compounds towards HCN addition is



- (1) $\text{(iii)} < \text{(i)} < \text{(iv)} < \text{(ii)}$
(2) $\text{(i)} < \text{(iii)} < \text{(iv)} < \text{(ii)}$
(3) $\text{(iii)} < \text{(iv)} < \text{(ii)} < \text{(i)}$
(4) $\text{(iii)} < \text{(iv)} < \text{(i)} < \text{(ii)}$

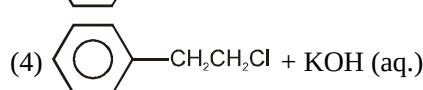
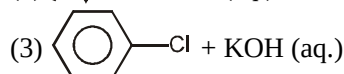
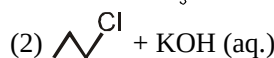
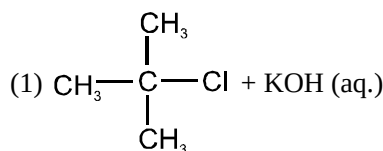
52. Which of the following compound not reacts with NaHSO_3 ?



53. Which among the following reactions forms acid anhydride?

- (1) R-COOH treated with $\text{NaHCO}_{3(\text{aq})}$
(2) R-COOH treated with PCl_5
(3) R-COOH (excess) heated in presence of P_2O_5
(4) R-COOH treated with R'-OH and heated

54. S_N^1 reaction is feasible in :

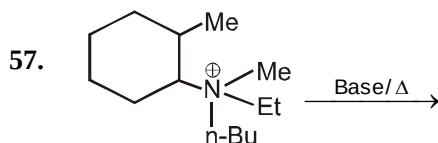


55. Elimination of HBr from B-bromobutane result in the formation of :

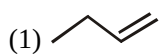
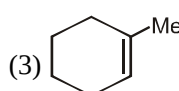
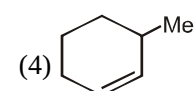
- (1) Predominantly B-butyne
(2) Predominantly 1-butene
(3) Predominantly 2-butene
(4) Equimolar mixture of 1 and 2-butene

56. Reaction of trans 2-phenyl -1 - bromocyclopentane on reaction with alcoholic KOH produces:

- (1) 2-phenylcyclopentene
- (2) 1-phenylcyclopentene
- (3) 3-phenylcyclopentene
- (4) 4-phenylcyclopentene

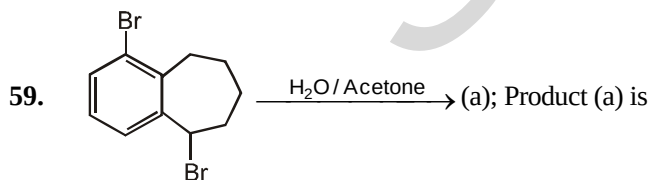


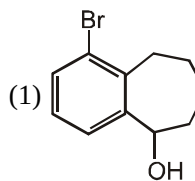
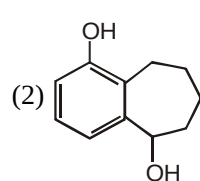
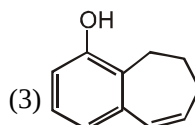
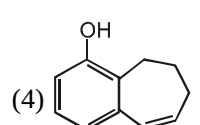
The alkene formed as a major product in the above elimination reaction is :

- (1) 
- (2) $\text{CH}_2 = \text{CH}_2$
- (3) 
- (4) 

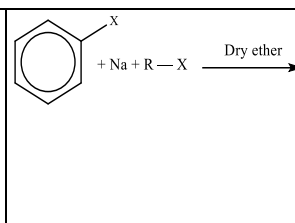
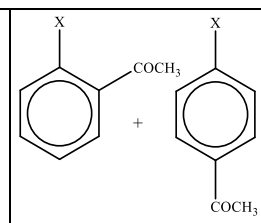
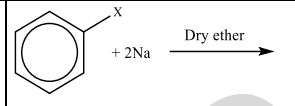
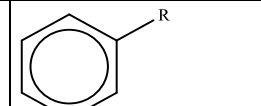
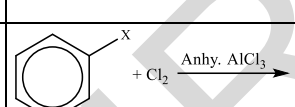
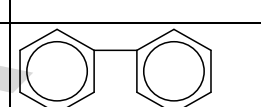
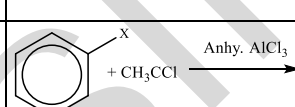
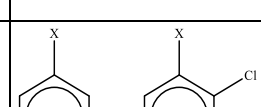
58. The increasing order of reactivity of the following isomeric halides with AgNO_3 (H_2O + alcohol) is

- (1) $\text{C}_6\text{H}_5 - \text{CH} = \text{CH} - \text{CH}_2 - \text{CH}_2 - \text{Cl}$
- (2) $\text{C}_6\text{H}_5 - \text{CH} = \text{CH} - \underset{\text{Cl}}{\text{CH}} - \text{CH}_3$
- (3) $\text{C}_6\text{H}_5 - \underset{\text{Cl}}{\text{C}} = \text{CH} - \text{CH}_2 - \text{CH}_3$
- (4) $\text{C}_6\text{H}_5 - \underset{\text{Cl}}{\text{C}}(\text{C}_6\text{H}_5) - \text{CH} = \text{CH}_2$



- (1) 
- (2) 
- (3) 
- (4) 

60. Match the following.

Column I (Reactant)		Column II (Product)	
A		p	
B		q	
C		r	
D		s	

- | | A | B | C | D |
|-----|---|---|---|---|
| (1) | r | s | q | p |
| (2) | q | r | s | p |
| (3) | q | p | r | s |
| (4) | p | q | r | s |

61. **Assertion:** p-dichlorobenzene has higher melting point than o-dichlorobenzene.

Reason: Stronger the van der Waal's forces of attraction, higher is the melting point.

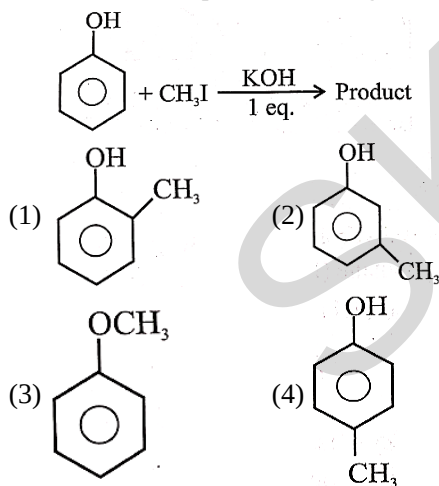
- (1) Both A and R are correct and R is correct explanation of A.
- (2) Both A and R are correct but R is not correct explanation of A.
- (3) A is correct but R is incorrect.
- (4) Both A and R are false.

62. Match the following conditions given in Column I with Column II and choose the correct option from the codes given below.

Column I		Column II	
A		P	NaOH, 443 K, H ⁺
B		Q	NaOH, 368 K, H ⁺
C		R	Warm H ₂ O

- (1) A → P; B → Q; C → R
 (2) A → R; B → Q; C → P
 (3) A → R; B → Q; C → P
 (4) A → Q; B → P; C → R

63. What is the final product of the given reaction?



64. Incorrect statement(s) is/are:

- (1) Iodoalkanes formed by Finkelstein reaction
 (2) Aryl halides cannot be prepared by replacement of hydroxyl group of phenol by halogen atom.
 (3) Treatment of chloroethane with a saturated solution of AgCN gives ethyl cyanide as the major product.
 (4) Both (1) and (3)

65. **Statement-I:** C₂H₅I is the least reactive alkyl halide.
Statement-II: C – I bond is the weakest and I[–] is a better leaving group than Br[–], Cl[–] & F[–] ions.
 (1) Both Statement-I and Statement-II are correct.
 (2) Both Statement-I and Statement-II are incorrect.
 (3) Statement-I is correct & Statement-II is incorrect.
 (4) Statement-I is incorrect & Statement-II is correct.

66. **Assertion:** The presence of nitro group facilitates nucleophilic substitution reactions in aryl halides.
Reason: The intermediate carbanion is stabilized due to the presence of nitro group.
 (1) If both assertion and reason are true and reason is the correct explanation of assertion.
 (2) If both assertion and reason are true but reason is not the correct explanation of assertion.
 (3) If assertion is true but reason is false.
 (4) If both assertion and reason are false.

67. Match the compounds given in Column-I with the effects given in Column-II.

Column-I		Column-II	
A	Chloramphenicol	I	Malaria
B	Thyroxine	II	Anaesthetic
C	Chloroquine	III	Typhoid fever
D	Chloroform	IV	Goitre
		V	Blood substituent

- (1) (A)-(I) (B)-(III) (C)-(IV) (D)-(II)
 (2) (A)-(III) (B)-(IV) (C)-(I) (D)-(II)
 (3) (A)-(II) (B)-(III) (C)-(IV) (D)-(I)
 (4) (A)-(III) (B)-(I) (C)-(II) (D)-(IV)

68. Match the species in Column I with corresponding reactants in Column II and choose the correct option.

Column I (Reaction)		Column II (Name)	
A	Gattermann reaction	1	C ₆ H ₅ COOAg · Br ₂ KCl ₄ / Heat
B	Hunsdiecker reaction	2	Hg ₂ F ₂ /Δ
C	Swarts reaction	3	C ₆ H ₅ N ₂ ⁺ Cl [–] , CuCl/HCl, Heat
D	Sandmeyer reaction	4	C ₆ H ₅ N ₂ ⁺ Cl [–] , Cu/HCl, Heat

- A B C D
 (1) 1 2 3 4
 (2) 4 1 2 3
 (3) 4 3 2 1
 (4) 1 2 4 3

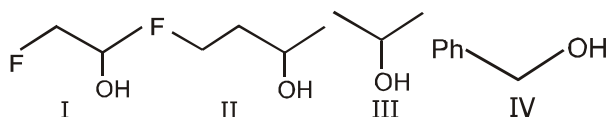
69. During dehydration of alcohols to alkenes by heating with concentrated H_2SO_4 the initiation step is

- (1) Protonation of alcohol molecule
- (2) Formation of carbocation
- (3) Elimination of water
- (4) Formation of an ester

70. Correct arrangement of the following nucleophiles in the order of their nucleophilic strength is

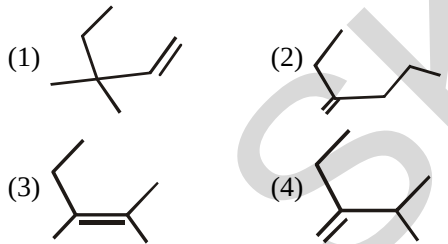
- (1) $\text{C}_6\text{H}_5\text{O}^- < \text{CH}_3\text{O}^- < \text{CH}_3\text{COO}^- < \text{OH}^-$
- (2) $\text{CH}_3\text{COO}^- < \text{C}_6\text{H}_5\text{O}^- < \text{CH}_3\text{O}^- < \text{OH}^-$
- (3) $\text{C}_6\text{H}_5\text{O}^- < \text{CH}_3\text{COO}^- < \text{CH}_3\text{O}^- < \text{OH}^-$
- (4) $\text{CH}_3\text{COO}^- < \text{C}_6\text{H}_5\text{O}^- < \text{OH}^- < \text{CH}_3\text{O}^-$

71. The correct order of reactivity of following alcohols towards conc. HCl/ZnCl_2 is:



- (1) $\text{I} > \text{II} > \text{III} > \text{IV}$
- (2) $\text{I} > \text{III} > \text{II} > \text{IV}$
- (3) $\text{IV} > \text{III} > \text{II} > \text{I}$
- (4) $\text{IV} > \text{III} > \text{I} > \text{II}$

72. $\xrightarrow{\text{Conc. H}_2\text{SO}_4}$ Major product is:

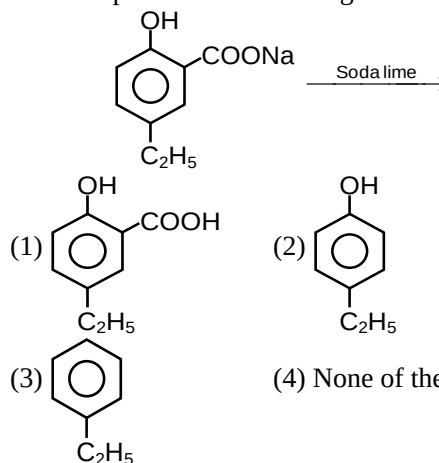


73. **Assertion:** The boiling point of alcohols are in the order t-butyl alcohol > s-butyl alcohol > n-butyl alcohol.

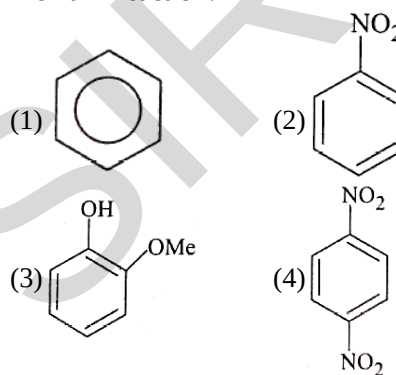
Reason: The difference between boiling points of two consecutive members of alcohols is about 50–60 K.

- (1) Both A and R are correct and R is correct explanation of A.
- (2) Both A and R are correct but R is not correct explanation of A.
- (3) A is correct but R is incorrect.
- (4) Both A and R are false.

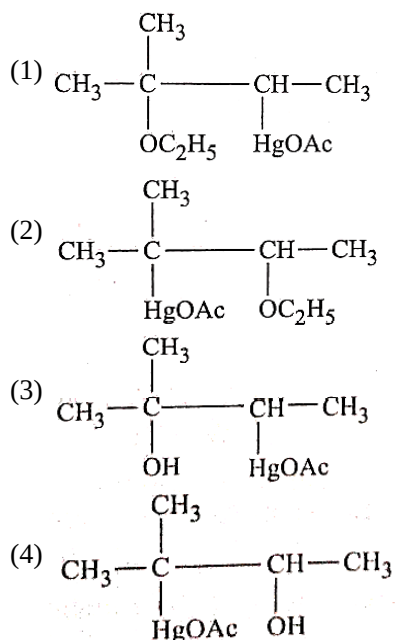
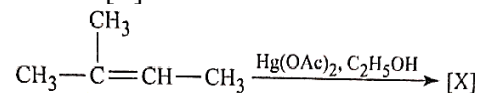
74. Give the product for following reaction.



75. Which of the following compound gives Reimer-Tiemann reaction?

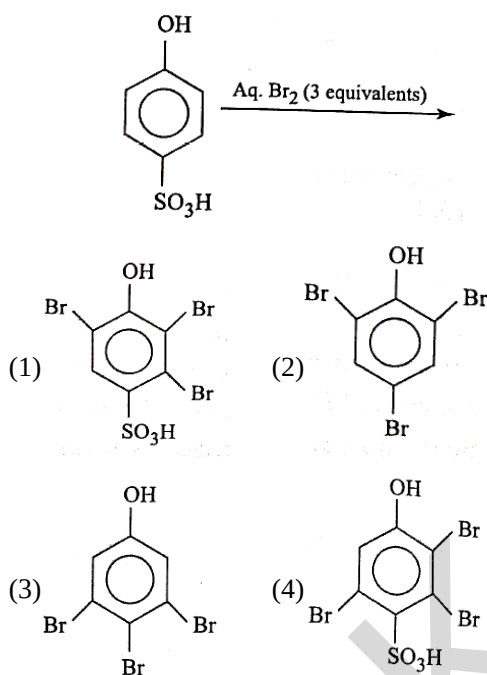


76. Product [X] of the reaction in

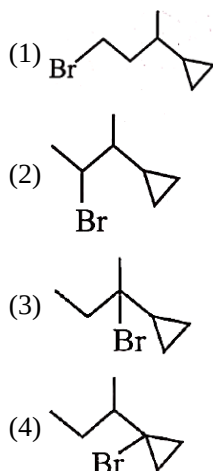


77. Phenol is a weaker acid than acetic acid because
- (1) Phenoxide ion is better stabilized by resonance than acetate ion.
 - (2) Acetate ion is better stabilized by resonance than phenoxide ion.
 - (3) Phenol is less soluble in water than acetic acid.
 - (4) Both phenoxide ion and acetate ion are stable.

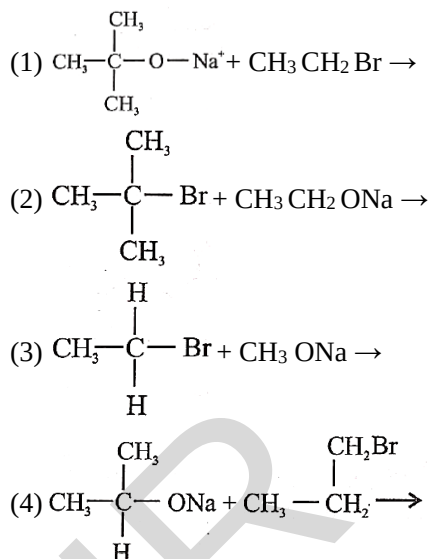
78. The major product of the following reaction is



79. product is:



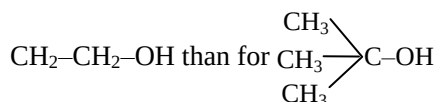
80. Which of the following reaction does not form ether as major product?



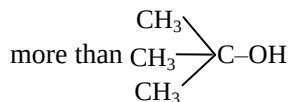
81. Which of the following statement is correct regarding chemical reactions of ethers?

- (1) The alkoxy group is only ortho directing and deactivates the aromatic ring towards electrophilic substitution reactions
- (2) The alkoxy group is meta directing and deactivates the aromatic ring towards electrophilic substitution reactions
- (3) The alkoxy group is ortho, para directing and activates the aromatic ring towards electrophilic substitution reactions
- (4) The alkoxy group is only para directing and deactivates the aromatic ring towards electrophilic substitution reactions

82. **Assertion:** Solubility in water is more for $\text{CH}_3\text{---CH}_2\text{---CH}_2\text{---CH}_2\text{---OH}$ than for $\text{CH}_3\text{---CH}_2\text{---CH}_2\text{---CH}_2\text{---OH}$



Reason: surface area of $\text{CH}_3\text{---CH}_2\text{---CH}_2\text{---CH}_2\text{---OH}$ is



- (1) If both (A) and (R) are true, and (R) is the correct explanation of (A).
- (2) If both (A) and (R) are true but (R) is not the correct explanation of (A).
- (3) If (A) is true but (R) is false.
- (4) If (A) is false but (R) is true.

83. **Statement I** : 3°alcohols show turbidity within 5 minutes, when treated with lucas reagent.

Statement II: Conc. HCl and anhydrous ZnCl_2 in 1 : 1 mixture is called Lucas reagent.

- (1) If both Statement-I and Statement-II are true, and Statement-II is the correct explanation of Statement-I.
- (2) If both Statement-I and Statement-II are true but Statement-II is not the correct explanation of Statement-I.
- (3) If Statement-I is true but Statement-II is false.
- (4) If Statement-I is false but Statement-I is true.

84. **Statement I** : Grignard reagent reacts with aldehydes and ketones to form alcohol.

Statement II: Alcohols have acidic hydrogen.

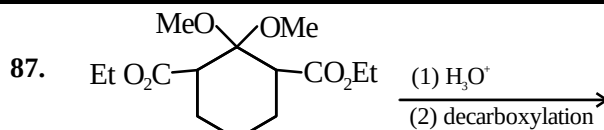
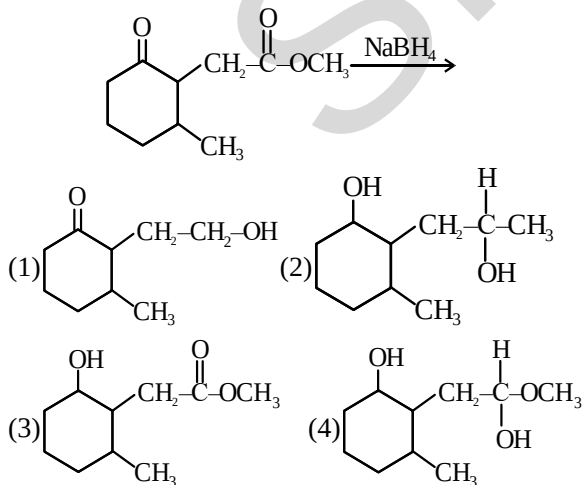
- (1) If both Statement-I and Statement-II are true, and Statement-II is the correct explanation of Statement-I.
- (2) If both Statement-I and Statement-II are true but Statement-II is not the correct explanation of Statement-I.
- (3) If Statement-I is true but Statement-II is false.
- (4) If Statement-I is false but Statement-I is true.

85. **Statement-I**: $\text{S}_{\text{N}}2$ reactions lead to Walden inversion.

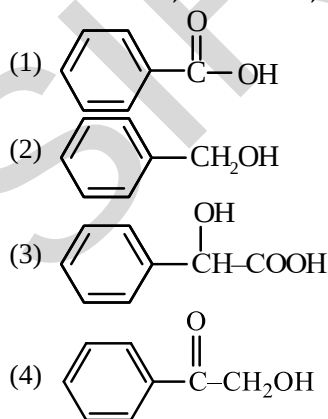
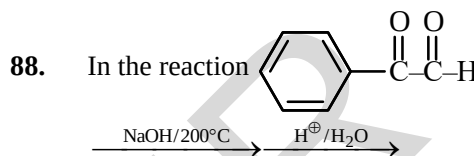
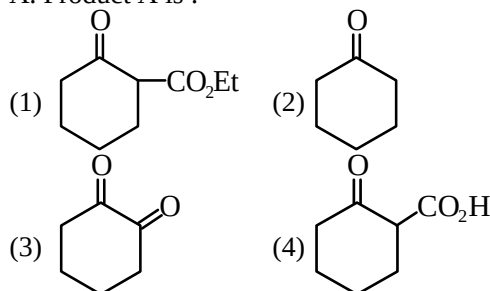
Statement-II: During transition state formation, nucleophile attacks the opposite side to that where halide is attached.

- (1) Both Statement-I and Statement-II are correct.
- (2) Both Statement-I and Statement-II are incorrect.
- (3) Statement-I is correct & Statement-II is incorrect.
- (4) Statement-I is incorrect & Statement-II is correct.

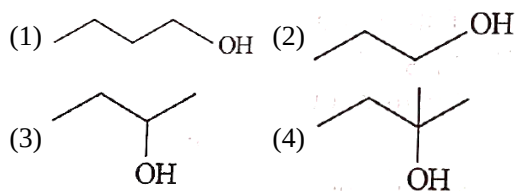
86. The product formed in the following chemical reaction is:



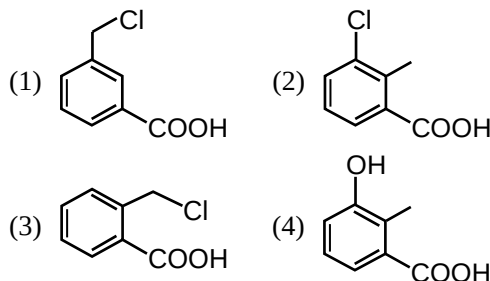
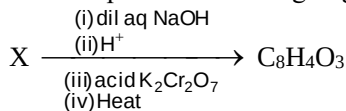
A. Product A is :



89. Which has maximum pK_{a} value?



90. The compound 'X' undergoing the following reaction is

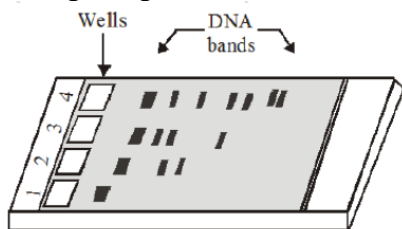


BIOLOGY

91. Bt toxin genes are isolated from *Bacillus thuringiensis* and incorporated into crop plants making them insecticidal. The choice of genes depends upon:

(1) Crop plant only
 (2) Targeted pest only
 (3) Both (1) & (2)
 (4) Neither crop nor targeted pest

92. A typical agarose gel electrophoresis showing migration of set of DNA fragments: Which of following statement is correct regarding this ?



(1) Only lane 1 shows undigested DNA fragment
 (2) Only lane 4 shows digested DNA fragment
 (3) Lane 1 to lane 3 show digested DNA
 (4) Only lane 4 shows undigested DNA Fragments

93. An antibiotic resistance gene in a vector which usually help in the selection of

(1) Transformed cells
 (2) Non-transformed cells
 (3) Competent cells
 (4) Non-recombinant cells

94. The critical research area of biotechnology is/are-

(1) Downstream processing technologies to purify the protein/organic compound.
 (2) Providing the best catalyst in the form of improved organism usually a microbe or pure enzyme.
 (3) Creating optimal conditions through engineering for a catalyst to act.
 (4) All of the above.

95. Genetically modified plants have been useful in increasing

(1) Crop yield
 (2) Nutritional value of food
 (3) Tolerance against abiotic stresses
 (4) All of the above

96. **Statement I:** In plants, the tumour inducing plasmid (Ti) of *Agrobacterium tumefaciens* is used as a cloning vector.

Statement II: Genetic modification has helped to reduce post harvest losses.

(1) Both statement I and II are correct
 (2) Both statement I and II are incorrect
 (3) Only statement I is correct
 (4) Only statement II is correct

97. Which gene was introduced in the first transgenic cow?

(1) Human α -lactalbumin
 (2) α -1-antitrypsin
 (3) β -1-antitrypsin
 (4) cry-IAc

98. Which Indian plants have either been patented or attempts have been made to patent them by Western nations for their commercial uses.

i. Basmati rice ii. Turmeric
 iii. Neem iv. Wheat
 (1) i, ii (2) ii, iii
 (3) i, ii, iii (4) i, ii, iii, iv

99. Choose the correct option regarding retrovirus:

(1) An RNA virus that can synthesis DNA during infection
 (2) A DNA virus that can synthesise RNA during infection
 (3) A ssDNA virus
 (4) A dsRNA virus

100. Statements related to human insulin are given below:

Which statement(s) is / are correct about genetically engineered insulin?

I. Pro-hormone insulin contain extra stretch of C-peptide
 II. A-peptide and B-peptide chains of insulin were produced separately in *E. coli*, extracted and combined by creating disulphide bond between them.
 III. Insulin used for treating Diabetes was extracted from Cattles and Pigs.
 IV. Pro-hormone insulin needs to be processed for converting into a mature and functional hormone.
 V. Some patients develop allergic reactions to the foreign insulin.

Choose the most appropriate answer from the options given below:

(1) II only (2) III and IV only
 (3) III, IV and V only (4) I, II and IV only

101. Main objective of production of pest resistant plants is:-

(1) Increase the nutritional value of food
 (2) Eliminate weeds from the field
 (3) Increase the amount of chemical pesticide used
 (4) Decrease the amount of chemical pesticide used

102. Use of biological resources of other countries without any legal authorization of the countries concerned is called

(1) Biopatent (2) Biopiracy
 (3) Bioethics (4) All of these

103. When DNA is transcribed into mRNA, usually the mRNA remains single-stranded, but in some cases an RNA can be made that is complementary to the mRNA. The is called ____ and its main function is to ____

- (1) Antisense RNA, block gene expression
- (2) Antisense RNA, amplify mRNA
- (3) Antisense RNA, enhance translation
- (4) Reverse transcription, enhance translation

104. Primers are

- (1) Small chemically synthesized oligonucleotides that are complementary to the region of DNA
- (2) Chemically synthesized oligonucleotides of about 10-18 nucleotides that are not complementary to the region of template DNA
- (3) The double-stranded DNA that need to the amplified
- (4) Specific sequences present on recombinant DNA

105. Match the following

Column-I		Column-II	
A	<i>Meloidogyne incognita</i>	i	Controls corn borer
B	<i>cryIAb</i>	ii	Controls bollworm
C	<i>cryIAc</i> and <i>cryIIAb</i>	iii	Gene therapy
D	ADA deficiency	iv	Infects roots of tobacco plant

- (1) A-(iv), B-(i), C-(ii), D-(iii)
- (2) A-(i), B-(ii), C-(iii), D-(iv)
- (3) A-(iv), B-(i), C-(iii), D-(ii)
- (4) A-(iv), B-(iii), C-(ii), D-(i)

106. Gel electrophoresis is used for

- (1) Cutting of DNA into fragments
- (2) Separation of DNA fragments according to their size
- (3) Construction of recombinant DNA by joining with cloning vectors
- (4) Isolation of DNA molecule

107. Match the following columns and select the correct option.

Column-I		Column-II	
A	α -1 antitrypsin	i	Separation & Purification
B	Down streaming process	ii	Cellular defence
C	RNAi	iii	Detection of HIV infection
D	PCR	iv	Emphysema

- (1) A-(iv), B-(i), C-(iii), D-(ii)
- (2) A-(i), B-(ii), C-(iii), D-(iv)
- (3) A-(iv), B-(ii), C-(iii), D-(i)
- (4) A-(iv), B-(i), C-(ii), D-(iii)

108. **Statement I-** *Plasmids* are *autonomously* replicating circular extrachromosomal DNA.

Statement II- Origin of replication is not a characteristic feature required by a cloning vector

- (1) Statement I is true, but statements II is false
- (2) Statement II is true, but Statement I is false
- (3) Statement I and Statement II are true
- (4) Statement I and Statement II are false

109. There are three basic steps in genetically modifying organism. Arrange these steps in correct sequence.

- A. Introduction of the identified DNA into the host.
- B. Maintenance of introduced DNA in the host and transfer of the DNA to its progeny.
- C. Identification of DNA with desirable genes.

- (1) A $\rightarrow$ B $\rightarrow$ C
- (2) B $\rightarrow$ C $\rightarrow$ A
- (3) C $\rightarrow$ B $\rightarrow$ A
- (4) C $\rightarrow$ A $\rightarrow$ B

110. In restriction enzymes (like *EcoRI*) the Roman numbers in the following name indicates the:

- (1) Order in which the enzyme were discovered from that strain of bacteria
- (2) Order in which the enzyme were isolated from that strain of bacteria.
- (3) Genus of the prokaryotic cell or bacteria
- (4) Strain of the bacteria.

111. Find the incorrect statement:

- (1) Genetic modification of organisms can have unpredictable results when organisms are introduced into the ecosystem.
- (2) There are an estimated 20,00,000 varieties of rich in India alone.
- (3) Bt toxin is a insecticide.
- (4) Golden rice is vitamin A enriched rice

112. Which of following is / are correct?

- I. Gene therapy is a method to correct gene defect that has been identified in an embryo or child.
- II. In gene therapy defective gene will be replaced by a normal, healthy and functional gene.
- III. A process by which a piece of RNA that is complementary sequence is used to stop expression of a specific gene is called as Antisense technology.
- IV. Bacteria genetically engineered to express a gene from a plant will not be able to synthesize a protein because translation is coupled with transcription and post transcriptional processing does not occur in it.

- (1) I, II only
- (2) II, III only
- (3) III, IV only
- (4) All of them

113. In pBR322, amp^R and tet^R genes are present. When we ligate a foreign DNA at *Sal-I* site, then recombinant plasmids will lose the resistance to the
 (1) Ampicillin (2) Tetracycline
 (3) Both (1) and (2) (4) None of the above
114. A typical bioreactor has
 A. An agitator system
 B. An oxygen delivery system
 C. A foam control system
 D. A temperature control system
 E. A pH control system
 F. Sampling ports
 (1) A, B, and C only
 (2) A, B, C and D only
 (3) A, B, C, D and E only
 (4) A, B, C, D, E and F
115. A stirred-tank bioreactor is usually cylindrical or with curved base to facilitate the
 (1) Mixing of the reactor contents
 (2) Control the temperature of the reactor
 (3) Control the pH of the reactor
 (4) All of the above
116. Which one of the following statement is not true regarding gel electrophoresis technique?
 (1) The separated DNA fragments are stained by using ethidium bromide.
 (2) The presence of chromogenic substrate gives blue coloured DNA bands on the gel.
 (3) Bright orange coloured bands of DNA can be observed in the gel when exposed to UV light.
 (4) The process of extraction of separated DNA strands from gel is called elution.
117. Stickiness of the ends of the DNA facilitates the
 (1) Action of DNA ligase and these ends joined together laterally.
 (2) Action of DNA ligase and these ends joined together end-to-end
 (3) Action of Taq polymerase
 (4) Action of restriction enzyme
118. **Statement I-** DNA being a hydrophilic molecule cannot pass through cell membranes
Statement II- The bacteria should be made competent to accept the DNA molecule
 The correct option regarding the above statements is
 (1) Statement I is true, but statements II is false
 (2) Statement II is true, but Statement I is false
 (3) Statement I and Statement II both are true
 (4) Statement I and Statement II both are false

119. In the yearA... the two enzymes responsible for ...B... the growth of bacteriophage in *E.coli* were isolated. One of these addedC.... group to DNA, while other ...D... DNA. The later was called ...E....
 (1) A-1967, B-promoting, C-ethyl, D-join, E-DNA ligase
 (2) A-1963, B-restricting, C-methyl, D-join, E-DNA ligase
 (3) A-1972, B-restricting, C-methyl, D-cut, E-restriction endonuclease
 (4) A-1963, B-restricting, C-methyl, D-cut, E-restriction endonuclease
120. Match the following columns.
- | Column-I | | Column-II | |
|----------|-----------------|-----------|----------------------|
| A | Biolistics | i | Animal cell |
| B | Micro-injection | ii | Bacterial cell |
| C | T-DNA | iii | Plant cell |
| D | Heat-shock | iv | <i>Agrobacterium</i> |
- (1) A-(ii), B-(iv), C-(i), D-(iii)
 (2) A-(iv), B-(iii), C-(ii), D-(i)
 (3) A-(i), B-(ii), C-(iii), D-(iv)
 (4) A-(iii), B-(i), C-(iv), D-(ii)
121. Now a days it is possible to detect the mutated gene causing cancer by allowing radioactive probe to hybridise its complimentary DNA in a clone of cells, followed by its detection using autoradiography because:
 (1) Mutated gene partially appears on a photographic film.
 (2) Mutated gene completely and clearly appears on a photographic film.
 (3) Mutated gene does not appears on a photographic film as the probe has no complementarity with it.
 (4) Mutated gene does not appear on photographic film as the probe has complementarity with it.
122. Now a days which of the following technique is used routinely to detect HIV infection ?
 (1) PCR (2) Serum analysis
 (3) Urine analysis (4) Northern blotting
123. Read the following sentences :
 (a) Nematode resistant plant developed by novel strategy based on RNA interference (RNAi)
 (b) RNAi involves silencing of specific mRNA due to complementary dsRNA
 (c) In RNAi, dsRNA prevents translation
 (d) RNAi is eukaryotic cellular defence mechanism
 Choose correct options :
 (1) a, b, c (2) b, c, d
 (3) a, b, d (4) a, b, c, d

124. What is true about the first case of constructing a recombinant DNA.
- It was performed in 1970
 - It was performed by Stanley Boyer and Herbert Cohen
 - It was a transfer of an antibiotic resistant gene
 - Gene was transferred from *Salmonella typhimurium* into *E.coli*
 - Gene could not replicate successfully in the target bacterium
- (1) I and V (2) I, II and III
(3) III and IV (4) III, IV and V

125. Which of the following shows correct chronological order of the events occurring during callus culture?
- Callus → Cell division → Explant → Addition of cytokinin → Cells acquire meristematic property
 - Explant → Callus → Cell division → Addition of cytokinin → Cells acquire meristematic property
 - Explants → Cell division → Callus → Addition of cytokinin → Cells acquire meristematic property
 - Callus → Explant → Cell division → Addition of cytokinin → Cells acquire meristematic property

126. **Assertion:** Restriction endonucleases are also called 'molecular scissors'
- Reason:** When fragments generated by restriction endonucleases are mixed, they join together due to their sticky ends
- Assertion is true, Reason is true; Reason is correct explanation for Assertion.
 - Assertion is true, Reason is true; Reason is not correct explanation for Assertion
 - Assertion is true, Reason is false
 - Assertion is false, Reason is true

127. Match the following columns

Column-I	Column-II
(A) Bam H 1	(i) Natural polymer extracted from sea weeds
(B) Bacteriophages	(ii) Introduction of DNA in a host bacterium
(C) Transformation	(iii) Virus infecting bacteria
(D) Agarose	(iv) tetR

- (1) A-(ii), B-(i), C-(iii), D-(iv)
(2) A-(iv), B-(iii), C-(ii), D-(i)
(3) A-(iii), B-(ii), C-(i), D-(iv)
(4) A-(i), B-(iv), C-(iii), D-(ii)

128. Select the correct match from the following

- Tobacco budworm – Coleopteran
- Armyworm – Lepidopteran
- Mosquito – Coleopteran
- Beetle – Dipteran

129. Select the correct answer about RNAi

- A.** Both sense and antisense RNA's being complementary to each other formed a double stranded (dsRNA) that terminates RNAi
- B.** It is followed by silencing of the specific rRNA
- (1) Only A is correct (2) Only B is correct
(3) Both are correct (4) None are correct

130. **Statement I:** Biotechnology deals with techniques that use living organism to produce products useful for humans.

Statement II: Construction of recombinant DNA involves cleaving DNA segments with endonuclease and rejoining with ligase.

- (1) Statement A is correct but Statement B is incorrect.
(2) Statement A is incorrect but Statement B is correct
(3) Both statements are correct.
(4) Both Statements are incorrect.

131. Match the following columns

Column-I	Column-II
A First transgenic cow	i mRNA silencing
B Safety test of polio vaccine	ii Golden Rice
C Resistance from nematode	iii Rosie
D Vitamin 'A' enriched crop	iv Transgenic mice

- (1) A-(i), B-(iii), C-(iv), D-(ii)
(2) A-(iii), B-(ii), C-(i), D-(iv)
(3) A-(iii), B-(iv), C-(ii), D-(i)
(4) A-(iii), B-(iv), C-(i), D-(ii)

132. **Statement I:** GMO help to reduce the reliance on chemical pesticides.

Statement II: GEAC makes decisions regarding the validity of GM research.

- (1) Both statement I and statement II are true
(2) Both statement I and statement II are false
(3) Statement I is true, but statement II is false
(4) Statement I is false, but statement II is true

133. Correctly match the column I with column II:

Column-I	Column-II
(A) Bt-cotton	(i) Growth hormone gene
(B) Nematode resistant tobacco	(ii) Human alpha lactalbumin
(C) Super mouse	(iii) Cotton bollworms
(D) Rosie	(iv) RNAi technique

- (1) A-iii, B-iv, C-i, D-ii
(2) A-iv, B-iii, C-i, D-ii
(3) A-ii, B-iv, C-i, D-iii
(4) A-iii, B-iv, C-ii, D-i

134. Identify incorrect pair :

- Restriction enzyme – Production of RFLP
- Reverse transcriptase – Production of cDNA from mRNA
- Taq polymerase – PCR
- DNA ligase – Production of sticky ends

- 135. Assertion:** Gene gun involves use of gold or tungsten pellets to transfer gene into plants cells.
Reason: *Agrobacterium* is used for vectorless gene transfer to animal cells.

- (1) If both assertion and reason are true and the reason is the correct explanation of the assertion.
- (2) If both assertion and reason are true, but reason is not the correct explanation of the assertion.
- (3) If assertion is true, but reason is false.
- (4) If both assertion and reason are false.

- 136.** Match the columns I and II.

Column-I		Column-II	
A	Genetic engineering	i	Permits variation
B	Bioprocess engineering	ii	Alter the chemistry of genetic material
C	Sexual reproduction	iii	Enabling growth of only the desired microbe

- (1) A-(iii), B-(p), C-(ii)
- (2) A-(ii), B-(iii), C-(i)
- (3) A-(i), B-(iii), C-(ii)
- (4) A-(ii), B-(i), C-(iii)

- 137.** In PCR technique which step requires thermostable enzyme :

- (1) Denaturation only
- (2) Annealing and extension
- (3) Only extension
- (4) Denaturation and extension

- 138. Assertion:** Restriction enzymes belong to a larger class of enzymes called nucleases.

Reason: Each restriction enzyme recognizes a specific palindromic nucleotide sequence in the DNA.

- (1) If both assertion and reason are true and the reason is the correct explanation of the assertion.
- (2) If both assertion and reason are true, but reason is not the correct explanation of the assertion.
- (3) If assertion is true, but reason is false.
- (4) If both assertion and reason are false.

- 139.** Which of the following is/are true?

- I. The Indian Parliament has recently cleared the second amendment of the Indian Patents Bill.
- II. Bioethics is the unauthorized use of bioresources and traditional knowledge related to bioresources for commercial benefits.
- III. Genome of many plants are manipulated or altered by combining them with other genes to get desired traits.

- (1) II only
- (2) I only
- (3) I and II
- (4) I and III

- 140.** For selectable marker.

- I. It helps to select the host cells which contain the vector and eliminate the non- transformants
- II. Genes encoding resistance to antibiotics like ampicillin, chloramphenicol, tetracycline or kanamycin, are useful selectable markers for *E.coli*

Which of the statements given above are correct?

- (1) Only I
- (2) Only II
- (3) I and II
- (4) None of these

- 141.** Palindrome sequences

- (1) Read same on two strands when orientation of reading is different.
- (2) Read different on two strands when orientation of reading is same.
- (3) Read same on two strands when orientation of reading is same.
- (4) Read opposite on two strands when orientation of reading is same.

- 142.** Electroporation procedure involves:

- (1) Fast passage of food through sieve pores in phloem elements with the help of electric stimulation.
- (2) Opening of stomatal pores during night by artificial light.
- (3) Making transient pores in the cell membrane to introduce gene constructs.
- (4) Purification of saline water with the help of a membrane system.

- 143.** Which of the following statement is correct in the context of observing DNA separated by agarose gel electrophoresis?

- (1) DNA can be seen in visible light
- (2) DNA can be seen without staining in visible light
- (3) Ethidium bromide stained DNA can be seen in visible light
- (4) Ethidium bromide stained DNA can be seen under exposure to UV light

- 144.** The techniques/processes that are included under biotechnology are

- (1) In vitro fertilisation
- (2) Correcting a defective gene
- (3) Developing DNA vaccine
- (4) All of these

- 145.** Consider the following statements about the responsibility of GEAC (set-up by the Indian Government)

- I. GEAC make decisions regarding the validity of the GM research.
- II. It checks the safety of introducing GM organisms for the public services for their only small scale use.

Which of the statements given above is/are correct?

- (1) Only I
- (2) Only II
- (3) I and II
- (4) None of these

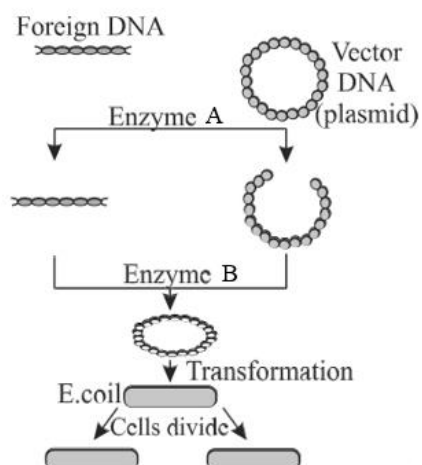
- 146.** In the naming of restriction enzymes the first letter is derived from ...A... name and next two letters from the ...B... and fourth letter from ...C... of ...D... where the enzymes are extracted

A to D in the statement can be

- (1) A- Genus, B-Species, C-Strain, D-Bacteria
- (2) A-Species, B-Genus, C-Strain, D-Bacteria
- (3) A-Genus, B-Species, C-Variety, D-Eukaryote
- (4) A-Species, B-Genus, C-Variety, D-Eukaryote

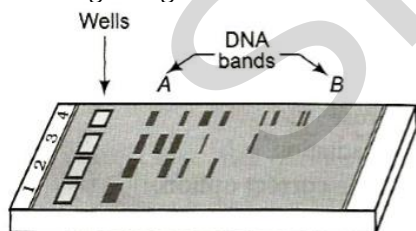
147. Consider the following statements about therapeutic drugs
- The recombinant DNA technology is used for production of therapeutic drugs which are safe and effective
 - It avoid unwanted immunological responses, commonly observed with similar products isolated from non-human sources
 - About thirty five recombinant therapeutics have been approved for human use in the world
- Which of the statements given above are correct?
- I and II only
 - I and III only
 - II and III only
 - I, II and III
148. How many fragments will be generated, if a closed circular DNA molecule is digested using a restriction enzyme having six recognition sites on the DNA?
- 12
 - 6
 - 7
 - 5
149. **Statement I-** Copy number is defined as the number of copies of plasmid present in a cell
Statement II- It varies from 15-100 copies per cell.
- Statement I is true, Statement II is false
 - Statement II is true, Statement I is false
 - Both statements are true
 - Both statements are false
150. In 1983, Eli Lilly an American company, first prepared two DNA sequences corresponding to A and B-chains of the human insulin and introduced them in the plasmids of *Escherichia coli* to produce insulin chains. Chains A and B were prepared separately, extracted and combined by creating
- Hydrogen bond
 - Disulphide bond
 - Covalent bond
 - Peptide bond
151. 'Restriction' in Restriction enzyme refers to:
- Cleaving of phosphodiester bond in DNA by the enzyme
 - Cutting of DNA at specific position only
 - Prevention of the multiplication of bacteriophage by the host bacteria
 - All of the above
152. Restriction endonuclease enzymes are used to cut
- Single stranded RNA
 - Double stranded DNA
 - Single stranded DNA
 - Double stranded RNA
153. Where would you find the smallest fragments of DNA after gel electrophoresis?
- Near the cathode, farthest away from the wells
 - Near the anode, close to the wells
 - Sometimes near the cathode, sometimes near the anode
 - Near the anode, farthest away to the wells
154. **Statement I:** Selectable markers help in identifying and eliminating non-transformants and selectively permitting the growth of transformants.
Statement II: If a recombinant DNA bearing gene for resistance to ampicillin is transferred into *E.coli* cells, the host cells become transformed into ampicillin resistant cells
- Both statement I and II are correct
 - Both statement I and II are incorrect
 - Only statement I is correct
 - Only statement II is correct
155. **Statement A:** Restriction enzymes *Hind II* is obtained from *Haemophilus*
Statement B: The first restriction endonuclease-Hind II
- Statement A is correct but Statement B is incorrect.
 - Statement A is incorrect but Statement B is correct.
 - Both statements are correct.
 - Both Statements are incorrect.
156. Number of H-bond in the recognition site of *EcoRI* is
- 6
 - 4
 - 14
 - 12
157. Which among the following be considered as a key tool in recombinant DNA technology
- Restriction enzyme
 - Polymerase enzymes
 - Ligases
 - Vectors
 - Host organisms
- d
 - e
 - c, e
 - All of these
158. Which of the following has the ability to transform normal cells into cancerous cells in animals?
- Retroviruses
 - Agrobacterium tumefaciens*
 - Plasmids
 - DNA viruses
159. Which one of the following is a correct statement?
- 'Bt' in 'Bt cotton' indicates that it is a genetically modified organism produced through biotechnology
 - Somatic hybridization involves fusion of two complete plant cells carrying desired genes
 - RNAi takes place in all eukaryotic organisms as a method of cellular defense
 - Over 95% of all existing transgenic animals are dogs
160. Which of the following option is true regarding Eco RI enzyme?
- Restriction endonuclease enzyme
 - Isolated from *Escherichia coli* RY13
 - Cuts at specific position within the DNA
 - GAATTC is its specific palindromic site
 - Obtained from a eukaryotic cell
- i, ii
 - i, ii, iii
 - i, ii, iii, iv
 - i, ii, iii, iv, v

161. Identify the enzyme (A) and (B) in the given diagram representing recombinant DNA technology.



- (1) Enzyme A- *Taq* Polymerase, Enzyme B- Restriction endonuclease
 (2) Enzyme A-Restriction endonuclease, Enzyme B- Ligases
 (3) Enzyme A- Ligases, Enzyme B-Restriction endonucleases
 (4) Enzyme A- Restriction endonuclease, Enzyme B- *Taq* Polymerase.
162. Choose the correct option about agrochemicals
- (1) These are expensive for farmers in developing countries
 (2) Also have harmful effects on environment
 (3) Genetically modified crops are less expensive than agrochemicals
 (4) All of the above

163. Study the given figure carefully and select the correct statements regarding this



- I. It represents typical agarose gel electrophoresis which showing differential migration of DNA fragments
 II. Lane 1 contains undigested DNA fragments
 III. Lanes 2 to 4 contains digested DNA fragment
 IV. Smallest DNA bands are present at (A) position and largest DNA bands are present at (B) position
- (1) I, II and III only (2) I, II and IV only
 (3) II and III only (4) III and IV only

164. Choose the correct options about Basmati rice
- I. In 1997, an American company got patent rights for Basmati rice through the US Patent and Trademark office and was allowed to sell a 'new variety' in US and abroad
 II. This new variety of Basmati was derived from Chinese farmer's varieties.
 III. Indian Basmati was crossed with semidwarf varieties and claimed as an invention or a novelty
- Which of the above statements are correct?
- (1) I and II only (2) I and III only
 (3) II and III only (4) I, II and III only

165. Consider the following statements.
- I. Transgenic animals are more sensitive to the toxic substance than non-transgenic animals
 II. Useful biological products can be produced by introducing into transgenic animals the portion of DNA which codes for a particular product
- Which of the statements are given above are correct?
- (1) I and II (2) Only II
 (3) I only (4) None of these

166. Identify the false statement
- (1) Transformation is a procedure through which a piece of DNA is introduced in a host bacterium
 (2) Fragments of the DNA can be separated by PCR
 (3) Exonuclease remove nucleotides from ends of the DNA.
 (4) Restriction endonuclease recognizes a specific sequence in the DNA called palindromic nucleotide sequence.

167. Match the following columns

Column-I		Column-II	
A	Plant cell wall	i	Pectinase
B	RNA	ii	Protease
C	Histone	iii	Cellulase
D	Pectine	iv	Ribonuclease

- (1) A-(ii), B-(i), C-(iii), D-(iv)
 (2) A-(i), B-(ii), C-(iv), D-(iii)
 (3) A-(iii), B-(iv), C-(i), D-(ii)
 (4) A-(iii), B-(iv), C-(ii), D-(i)

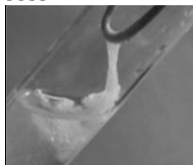
168. After ligation of foreign DNA at the Pst I site the recombinant plasmids will
- (1) Be able to translate erythromycin resistant gene
 (2) Not be able to translate tetracycline resistant gene
 (3) Be able to translate ampicillin resistant gene
 (4) Not be able to translate ampicillin resistant gene
169. A bacterial cell was transformed with a recombinant DNA molecule that was generated using a human gene. However, the transformed cells did not produce the desired protein. Reasons could be:
- (1) Human gene may have intron which bacteria cannot process
 (2) Amino acid codons for humans and bacteria are different
 (3) Human protein is formed but degraded by bacteria
 (4) All of the above

170. Consider the given statements w.r.t. gel electrophoresis.
- DNA can be separated by forcing them to move towards anode under an electric field.
 - DNA fragments separate according to their size through sieving effect provided by agarose gel.
 - The used matrix in this technique is agarose which is an artificial polymer.

Which of the above statements is/ are correct?

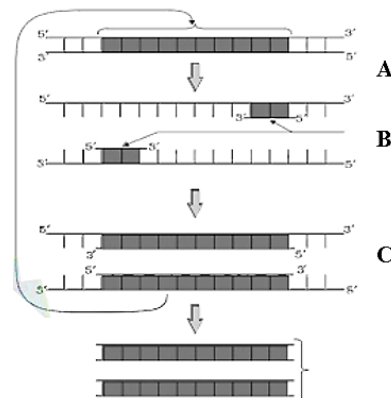
- (1) II only (2) I and II
 - (3) I, II, and III (4) II and III
171. i. In elution the separated bands of DNA are cut out from agarose gel and extracted from the gel piece.
 ii. *E. coli* cloning vector shows several restriction site, Ori, antibiotic resistance genes and Rop.
 iii. The down stream processing and quality control testing vary from products to product.
 iv. Competent bacterial cell cannot take up the plasmid.
- (1) All are incorrect (2) All are correct
 - (3) Only iv is correct (4) Except iv all are correct
172. Choose correct statement
- (1) If a protein encoding gene is expressed in a heterologous host it is called Recombinant protein
 - (2) Ligation of alien DNA is carried out at a restriction site present in Ori
 - (3) In pBR322 one antibiotic resistance gene helps in selecting the transformants, whereas the other helps in cloning
 - (4) Rop codes for proteins involved in ligation of foreign DNA
173. Which of the following method can be used for making the bacterial cell competent?
- (1) Treating with specific concentration of monovalent cation.
 - (2) Treating with specific concentration of divalent cation
 - (3) Heat shock + ice treatment
 - (4) Both (2) & (3)
174. Which step proved to be the main challenging obstacle in the production of human insulin by genetic engineering?
- (1) Splitting A & B polypeptide chains
 - (2) Addition of C-peptides to pro-insulin
 - (3) Getting insulin assembled into mature form
 - (4) Removal of C-peptide from active insulin

175. Identify the process



- (1) Elution (2) Spooling
- (3) Spooning (4) Sieving

176. Identify the processes labelled a, b, c in the following diagram.



- (1) A - annealing, B-denaturation, C- extension
 - (2) A-extension B-denaturation, C - annealing
 - (3) A- denaturation, B-Annealing, C - Extension
 - (4) A- annealing, B-extension, C- denaturation
177. **Assertion (A):** In the process of transformation, all cells take up the foreign DNA.
Reason (R): Cloning sites help to identify the cells that have taken up the foreign DNA.
- (1) Both Assertion and Reason are the true, and reason is a correct explanation of assertion.
 - (2) Both Assertion and Reason are the true, but reason is not a correct explanation of assertion.
 - (3) Assertion is true, but reason is false.
 - (4) Assertion and reason both are false.
178. **Statement:-** pBR322 is a cloning vector in fungi
Statement II: Lysozymes are used to break the bacterial cell wall to obtain the DNA.
- (1) Statement I and statement II both are correct
 - (2) Statement I is correct, but statement II is incorrect.
 - (3) Statement I is incorrect, but statement II is correct.
 - (4) Statement I and statement II both are incorrect.
179. Which of the following statements is incorrect?
- (1) Meristem culture can help in recovering healthy plants from diseased one.
 - (2) Isolated protoplasts can be fused to form hybrid protoplasts.
 - (3) Somatic hybridization is a process used to fuse animal cells.
 - (4) The hybrid plant pomato was created by fusing protoplasts of tomato and potato.
180. Which statement correctly describes the recent (second) amendment by the Indian Parliament?
- (1) The amendment disregards issues related to patent terms and research initiatives.
 - (2) The Indian Patents Bill amendment considers issues such as patent terms, emergency provisions, and research and development initiatives.
 - (3) The amendment exclusively benefits foreign multinational companies.
 - (4) The amendment was rejected by the Indian parliament.

Answer Key and Solution

ANSWER KEY & SOLUTION

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	2	3	3	3	2	1	1	3	2	3	4	4	3	1	3
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	3	3	2	1	1	3	1	2	3	4	1	3	1	4	4
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	4	1	1	3	4	3	4	3	3	2	3	1	4	3	1
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	2	4	4	3	4	1	4	3	1	3	3	2	3	1	2
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	2	1	3	3	4	1	2	2	1	4	3	3	4	2	3
Que.	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Ans.	1	2	2	3	2	3	4	4	2	1	3	2	3	4	3
Que.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Ans.	3	1	1	4	4	1	1	3	1	1	4	2	1	1	1
Que.	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Ans.	2	4	1	4	2	2	4	2	4	1	2	2	3	4	4
Que.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135
Ans.	3	1	4	3	3	2	2	2	4	3	4	1	1	4	3
Que.	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
Ans.	2	3	2	4	3	3	3	4	4	1	1	4	2	2	2
Que.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165
Ans.	3	2	4	1	3	4	4	1	3	3	2	4	1	2	1
Que.	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
Ans.	2	4	4	1	2	4	1	4	3	2	3	4	3	3	2

PHYSICS

1. (2)

Sol. Relative velocity of image w.r.t. man
 $= 15 - (-15) = 30 \text{ m/s}$

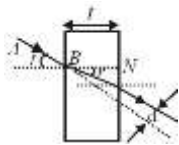


2. (3)

Sol.

3. (3)

Sol.



From figure, in right angled $\triangle CDB$

$$\angle CBD = (i - r)$$

$$\therefore \sin(i - r) = \frac{CD}{BC} = \frac{d}{BC}$$

$$\text{or } d = BC \sin(i - r) \dots (i)$$

Also, in right angled $\triangle CNB$

$$\cos r = \frac{BN}{BC} = \frac{t}{BC}$$

$$\text{or } BC = \frac{t}{\cos r} \dots (ii)$$

Substitution equation (ii) in equation (i), we get

$$d = \frac{t}{\cos r} \sin(i - r)$$

For small angles $\sin(i - r) \approx \sin i - r \cos r \approx 1$

$$d = t(i - r), d = it \left[1 - \frac{r}{i} \right]$$

4. (3)

Sol. For the prism as the angle of incidence (i) increases, the angle of deviation (δ) first decreases goes to minimum value and then increases.

5. (2)

Sol.
$$\beta = \frac{\lambda D}{d} = \frac{6000 \times 10^{-10} \times 2}{4 \times 10^{-3}} = 0.3 \times 10^{-3} \text{ m} = 0.3 \text{ mm}$$

6. (1)

Sol. Let distance = u . Now $\frac{v}{u} = 16$ and $v = u + 120$

$$\therefore \frac{120 + u}{u} = 16 \Rightarrow 15u = 120 \Rightarrow u = 8 \text{ cm.}$$

7. (1)

Sol. Real, inverted and same in size because object is at the centre of curvature of the mirror.

8. (3)

$$\mu \propto \frac{1}{\lambda} \Rightarrow \frac{\mu_1}{\mu_2} = \frac{\lambda_2}{\lambda_1} \Rightarrow \frac{1}{\frac{4}{3}} = \frac{\lambda_2}{4200}$$

$$\Rightarrow \lambda_2 = 3150 \text{ \AA}$$

9. (2)

Sol. For minimum deviation, incident angle is equal to emerging angle.

$\therefore$ QR is horizontal.

10. (3)

Sol. Here focal length = f and $u = -f$

On putting these value in $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

$$\Rightarrow \frac{1}{f} = -\frac{1}{f} + \frac{1}{v} \Rightarrow v = \frac{f}{2}$$

11. (4)

Sol. Image formed by convex mirror is always. Erect diminished and virtual.

12. (4)

Sol.

13. (3)

Sol. Fringe width is $\beta = \frac{D\lambda}{d}$ where D is the distance of the slits from the screen, d is the separation of the slits and λ , the wavelength.

14. (1)

$$\text{By using } \frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} \Rightarrow \frac{1}{F} = \frac{1}{+40} + \frac{1}{-25}$$

$$\Rightarrow F = -\frac{200}{3} \text{ cm,}$$

$$\text{hence } P = \frac{100}{f(\text{cm})} = \frac{100}{-200/3} = -1.5 \text{ D}$$

15. (3)

Sol.
$$\lambda \propto \frac{1}{\mu} \Rightarrow \frac{\lambda_1}{\lambda_2} = \frac{\mu_2}{\mu_1} = \frac{\mu}{1}$$

16. (3)

Sol.
$$\mu = \frac{\sin i}{\sin r} = \frac{V_{\text{air}}}{0.75 V_{\text{air}}} = \frac{100}{75} = \frac{4}{3}$$

$$= \frac{i}{r} = \frac{4}{3} \Rightarrow r = \frac{3i}{4}, \delta = i - r$$

$$= i - \frac{3i}{4} = \frac{i}{4}$$

17. (3)

Sol.
$$n_1 \lambda_1 = n_2 \lambda_2 \Rightarrow n(7.8 \times 10^{-5}) = (n+1)(5.2 \times 10^{-5})$$

 or $n(2.6 \times 10^{-5}) = 5.2 \times 10^{-5}; \therefore n = 2$

18. (2)

Sol.

19. (1)

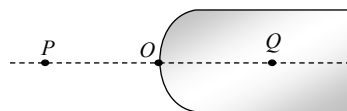
$$\text{By using } \frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

Where $\mu_1 = 1$, $\mu_2 = 1.5$, $u = -OP$, $v = OQ$

$$\text{Hence } \frac{1.5}{OQ} - \frac{1}{-OP} = \frac{1.5 - 1}{(+R)} \Rightarrow$$

$$\frac{1.5}{OP} + \frac{1}{OP} = \frac{0.5}{R}$$

$$\Rightarrow OP = 5R$$



20. (1)

Sol. In Young's double slit experiment intensity at a point is given by

$$I = I_0 \cos^2 \left(\frac{\phi}{2} \right)$$

$$\text{or } \frac{I}{I_0} = \cos^2 \left(\frac{\phi}{2} \right) \quad \dots(i)$$

Phase difference, $\phi = \frac{2\pi}{\lambda} \times \text{path difference}$

$$\therefore \phi = \frac{2\pi}{\lambda} \times \frac{\lambda}{6} \text{ or } \phi = \frac{\pi}{3} \quad \dots(ii)$$

Substitute eqn. (ii) in eqn. (i), we get

$$\frac{I}{I_0} = \cos^2 \left(\frac{\pi}{6} \right) \text{ or } \frac{I}{I_0} = \frac{3}{4}$$

21. (3)

Sol.

22. (1)

$$\text{Sol. } \mu_m = \frac{c}{v} = \frac{n\lambda_a}{n\lambda_m} = \frac{\lambda_a}{\lambda_m}$$

23. (2)

Sol. Applying the lens maker's formula

$$\frac{1}{f} = P = (n-1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

We know that the greater positive power is for that material for which $\frac{1}{R_1} - \frac{1}{R_2}$ is maximum and positive.

24. (3)

$$\text{Sol. Fringe width, } \beta = \frac{\lambda D}{d}$$

where λ is the wavelength of light, D is the distance between screen and the slits and d is the distance between two slits

$$\therefore \Delta\beta = \frac{\lambda}{d} \Delta D \quad (\text{As } \lambda \text{ and } d \text{ are constants})$$

$$\text{or } \lambda = \frac{\Delta\beta d}{\Delta D}$$

Substituting the given values, we get

$$\lambda = \frac{(3 \times 10^{-5} \text{ m})(10^{-3} \text{ m})}{(5 \times 10^{-2} \text{ m})} = 6 \times 10^{-7} \text{ m} = 6000 \text{ \AA}$$

25. (4)

Sol. As $r_1 < i_1$ i.e., the incident ray bends towards the normal $\Rightarrow$ normal 2 is denser than medium 1.

Or $r_2 < r_1 \Rightarrow$ Medium 3 is denser than medium 1.

Also, $r_2 > r_1 \Rightarrow$ medium 2 is denser than medium 3.

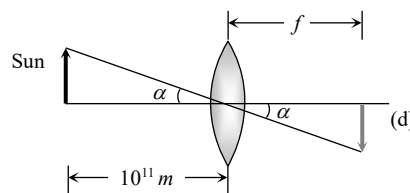
26. (1)

Sol.

27. (3)

From figure

$$\frac{D}{d} = \frac{10^{11}}{2} \Rightarrow d = \frac{2 \times 1.4 \times 10^9}{10^{11}} = 2.8 \text{ cm.}$$



28. (1)

$\delta = (\mu - 1)A$ as A is halved, so δ is also halved

29. (4)

Sol. The amplitude will be $A \cos 60^\circ = A/2$

30. (4)

Sol.

31. (4)

Sol.

32. (1)

Sol

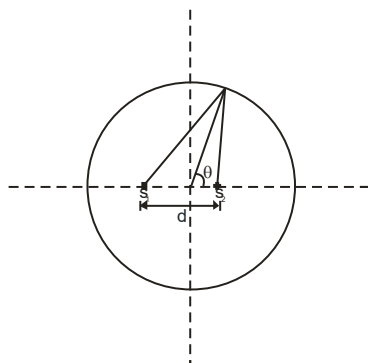
33. (1)

$$\text{Sol. } \frac{1}{f} = \frac{1}{v} + \frac{1}{u}, u = -x, \frac{v}{u} = \frac{1}{n}$$

$$v = +\frac{x}{n}$$

$$\frac{n-1}{x} = \frac{1}{f}$$

34. (3)



$$\Delta x = d \cos \theta = n\lambda$$

$$d = 2\lambda$$

$$\therefore \frac{n}{2}$$

$$n = 1$$

$$\cos \theta = \frac{1}{2}$$

$$\theta = 60^\circ$$

35. (4)

$$\text{Sol. } \frac{I_{\max}}{I_{\min}} = \frac{(a+b)^2}{(a-b)^2} = \frac{36}{1} \Rightarrow \frac{a+b}{a-b} = \frac{6}{1}$$

$$a + b = 6a - 6b \Rightarrow 5a \quad \text{or} \quad \frac{a}{b} = \frac{7}{5}$$

36. (3)

Sol. In interference of light the energy is transferred from the region of destructive interference to the region of constructive interference. The average energy being always equal to the sum of the energies of the interfering waves. Thus the phenomenon of interference is in complete agreement with the law of conservation of energy.

37. (4)

Sol. Velocity and wavelength change but frequency remains same.

38. (3)

$$\text{Sol. } a \sin 30^\circ = \lambda \Rightarrow a = 2\lambda$$

39. (3)

Sol.

40. (2)

Sol. $\tan i = \mu$ (Snell's Law)

41. (3)

Sol. $\because i_p = \phi$, therefore, angle between reflected and refracted rays is 90°

42. (1)

Sol. According to Cartesian sign convention

Object distance, $u = -15$ cm

Focal length, $f = -10$ cm

Using mirror formula

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f} \Rightarrow \frac{1}{(-15)} + \frac{1}{v} = \frac{1}{(-10)}$$

$$\frac{1}{v} = \frac{1}{(-10)} - \frac{1}{(-15)} = \frac{1}{(-10)} + \frac{1}{(-15)} \quad \text{or} \quad v = -30 \text{ cm}$$

The image is 30 cm from the mirror on the same side of the object.

$$\text{Magnification, } m = -\frac{v}{u} = -\frac{(-30\text{cm})}{(-15\text{cm})} = -2\text{cm}$$

The image magnified, real and inverted.

43. (4)

It will be concentric circles

44. (3)

Sol. Here, $A_1 = A$, $A_2 = 2A$, $\phi = 60^\circ$

Amplitude of the resultant wave, $A_R =$

$$\sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \phi}$$

$$= \sqrt{A^2 + (2A)^2 + 2 \times A \times 2A \times \cos 60^\circ} = \sqrt{7A^2}$$

Intensity $\propto$ (Amplitude) 2

$$\therefore \text{Intensity} \propto A_R^2 = 7A^2$$

45. (1)

CHEMISTRY

46. (2)

Sol: A = propan-2-ol, C = propane-1-ol

47. (4)

Sol: Carbocyclic acid having at least one α -H, will give HVZ reaction.

48. (4)

Sol:

- +ve DNP test indicating that compound (A) must have carbonyl group
- +ve iodoform test indicating that (A) must be methyl ketone

49. (3)

Sol: $\Rightarrow$ Rate of nucleophilic addition reaction $\propto$ $\frac{1}{\text{ERG}}$
 $\Rightarrow$ Ketone always less reactive for N.A.R due to steric hindrance, II > III > I

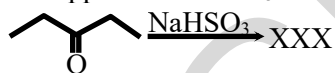
50. (4)

Sol: Contain α -H

51. (1)

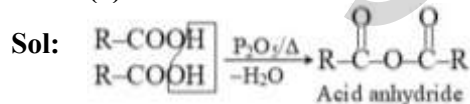
Sol: -I, -M effect of NO_2 increase reactivity towards nucleophilic addition reaction with HCN.

52. (4)

Sol: Aldehyde and only aliphatic methyl ketone gives white ppt with NaHSO_3 

ethyl ketone

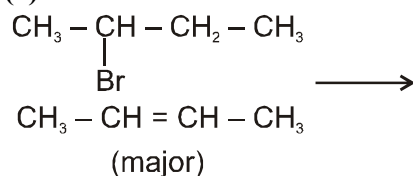
53. (3)



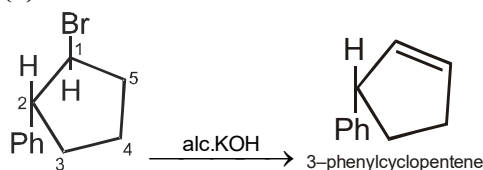
54. (1)

 S_N^1 reaction $\rightarrow$ stability of carbocation

55. (3)

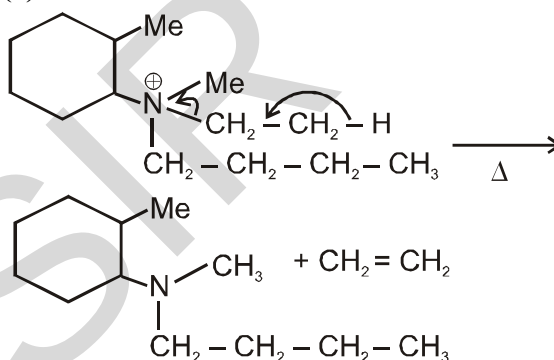


56. (3)



It is anti elimination reaction so hydrogen atom from second carbon will not be eliminated as it is in syn-position rather hydrogen atom from 5th carbon will be eliminated.

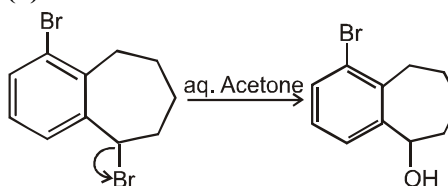
57. (2)



58. (3)

On the basis of carbocation stability.

59. (1)



In aryl halides the C-X bond has partial double bond character due to resonance so it will not give S_N reaction.

60. (2)

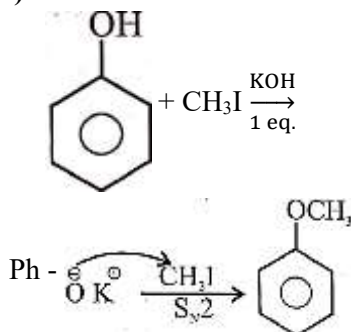
61. (2)

Among dichlorobenzenes, the p-isomer being symmetrical packs closely in the crystal lattice and hence, has much higher melting point than o- and m-isomers.

62. (1)

The presence of EWG ($-\text{NO}_2$) increases the NSR tendency on benzene. Thus, as the number of $-\text{NO}_2$ group increases, the reaction gets favourable even in normal condition.

63. (3)



64. (3)

Treatment of chloroethane with a saturated solution of AgCN gives ethyl isocyanide as the major product. AgCN is mainly covalent in nature and acts as a nucleophile through lone pair of electrons present on N to give ethyl isocyanide.

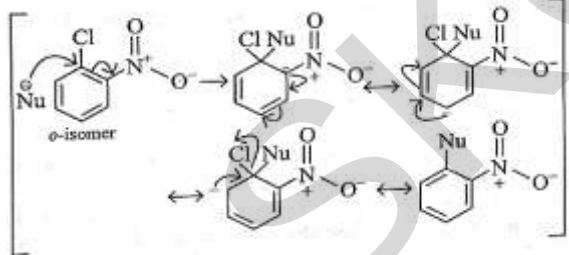
65. (4)

Since, C-I bond is the weakest and I^- is a better leaving group than Br^- , Cl^- and F^- ions, hence $\text{C}_2\text{H}_5\text{I}$ is the most reactive alkyl halide.

66. (1)

The C-atom of C-Cl in chloronitrobenzene carries a small positive charge.

Because of the positive charge, attack at this carbon atom by the nucleophile is facilitated.

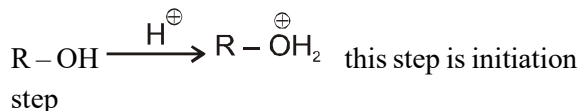


67. (2)

68. (2)

The correct match is A-4, B-1, C-2, D-3.

69. (1)



70. (4)

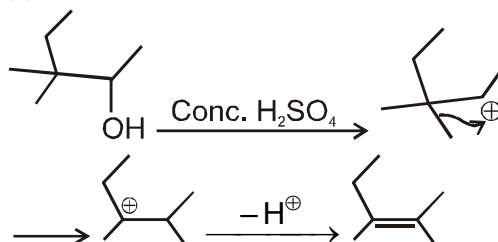
As increases delocalisation of negative charge nucleophilicity decreases.

71. (3)

According to carbocation stability



72. (3)

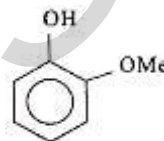


73. (4)

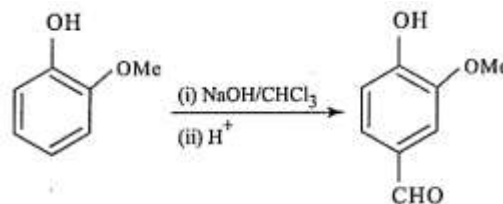
Branching in organic molecules give rise to a decrease in surface area and thus, intermolecular forces among the molecules decrease to show low boiling points. The difference between boiling points of two consecutive members of alcohols is about 18–20 K.

74. (2)

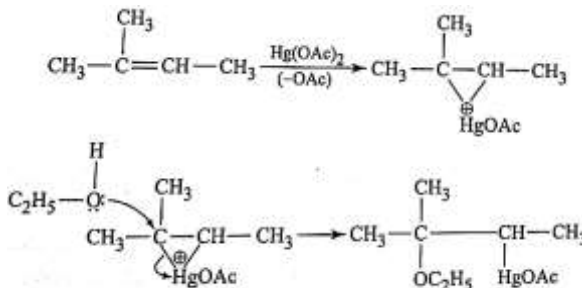
75. (3)



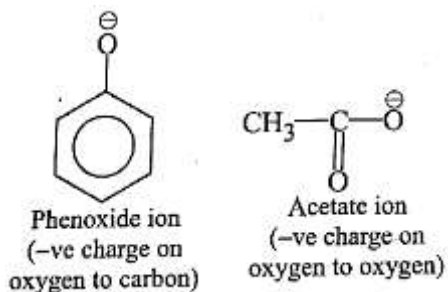
will give Reimer-Tiemann reaction in which the phenol group loses its proton in alkaline medium to form phenoxide ion which activate the ring so it can easily attack on carbene.



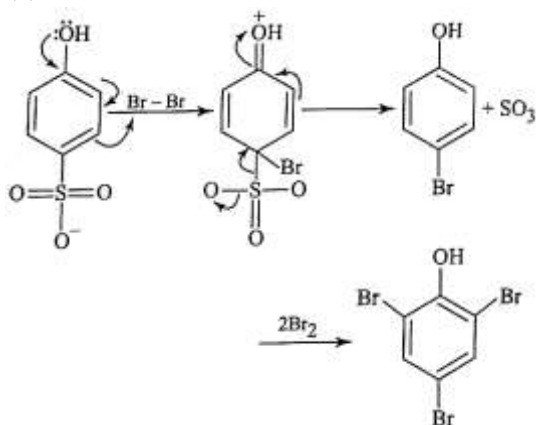
76. (1)



77. (2)

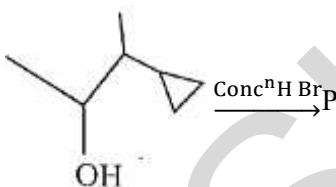


78. (2)

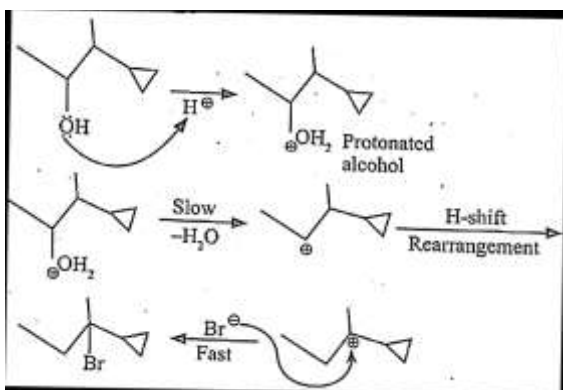
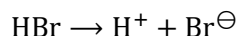


79. (3)

It is a S_N1 reaction, S_N1 rearrangement takes place in carbocation

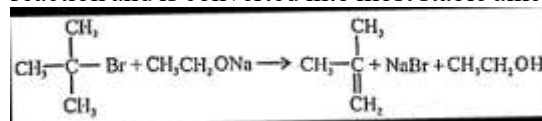


Mechanism:- The given alcohol is a secondary alcohol, therefore HBr will react here without catalyst. The reaction will proceed via S_N1 mechanism.



80. (2)

Tertiary alkyl bromide gives alkene as a major product in presence of sodium ethoxide because tertiary carbocation readily gives elimination reaction and is converted into most stable alkene.



81. (3)

The alkoxy group in ethers is ortho, para directing in nature and activates the aromatic ring towards electrophilic substitution reactions.

82. (4)

83. (4)

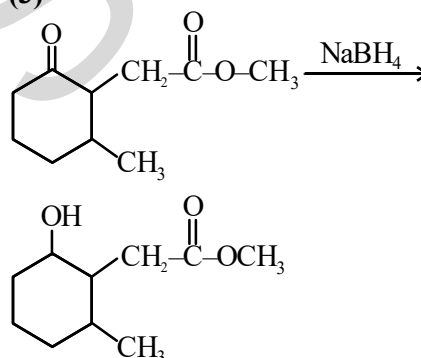
84. (2)

85. (1)

Stereo chemical inversion of a molecule is known as Walden inversion i.e., chiral center inversion in a molecule in chemical reaction.

i.e. $[\text{NO}_2^- < \text{NO}_2 < \text{NO}_2^+]$

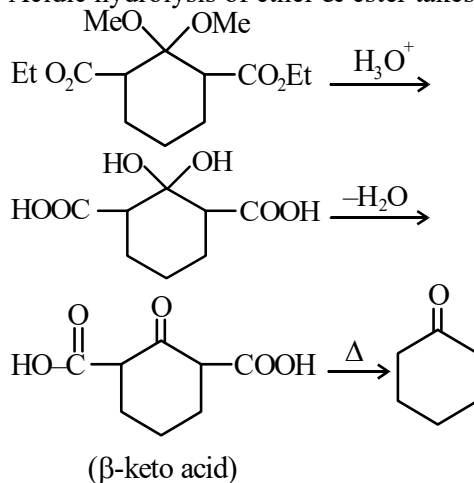
86. (3)



NaBH_4 reduces aldehyde/ketone but does not reduce ester.

87. (2)

Acidic hydrolysis of ether & ester takes place



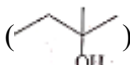
88. (3)
Intramolecular Cannizzarro's reaction

89. (4)

Sol. $pK_a = -\log K_a$

Acidic character $\propto K_a \propto \frac{1}{pK_a}$

1° alcohol () is the strongest acid thus has maximum K_a value and

minimum pK_a value 3° alcohol () is

the weakest acid and thus has minimum K_a value and maximum pK_{a_1} value.

90. (3)

